STRONG GENERATOR FOR PERFECT COMPLEXES ON SMOOTH PROPER SCHEMES

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Diagonal trick. lemma-diagonal-trick L26475 (carded this batch). Provides E and G with O_Delta in the thick hull of E boxtimes G.
- smad machinery. derived-section-operate-on-full, derived-lemma-find-smallest-containing-E, derived-remark-operations-functor (cross-chapter, named). Definitions and the functoriality of smd(add(...)).
- Cohomology base change. perfect-lemma-cohomology-base-change (cross-chapter, named). Computes the pushforward of the box product.

Source reference. Original reference 0ATG: focused-seeds/geometry/S9-eqv-smooth-proper-strong-generator/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FX5: focused-seeds/geometry/S9-eqv-smooth-proper-strong-generator/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FX0: focused-seeds/geometry/S9-eqv-smooth-proper-strong-generator/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0FYV: sources/stacks-equiv/pinned-original-source.tex, source lines 1455–1455. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FZ5: sources/stacks-equiv/pinned-original-source.tex, source lines 1942–1942. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 08EU: focused-seeds/geometry/S9-eqv-smooth-proper-strong-generator/ref-tags.txt, source lines 8–8. Navigation pointer only: the

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counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be a scheme proper and smooth over k. Then $D_{perf}(\mathcal{O}_X)$ has a strong generator.*

Proof. Using Lemma 0FZ5 choose finite locally free $\mathcal{O}_X$-modules $\mathcal{E}$ and $\mathcal{G}$ such that $\mathcal{O}_\Delta \in \langle \mathcal{E} \boxtimes \mathcal{G} \rangle$ in $D(\mathcal{O}_{X\times X})$. We claim that $\mathcal{G}$ is a strong generator for $D_{perf}(\mathcal{O}_X)$. With notation as in Derived Categories, Section 0FX0 choose $m, n \geq 1$ such that

$$\mathcal{O}_\Delta \in smd(add(\mathcal{E} \boxtimes \mathcal{G}[-m, m])^{\star n})$$

This is possible by Derived Categories, Lemma 0ATG. Let K be an object of $D_{perf}(\mathcal{O}_X)$. Since $L\mathrm{pr}_1^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} -$ is an exact functor and since

$$L\mathrm{pr}_1^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} (\mathcal{E} \boxtimes \mathcal{G}) = (K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{E}) \boxtimes \mathcal{G}$$

we conclude from Derived Categories, Remark 0FX5 that

$$L\mathrm{pr}_1^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} \mathcal{O}_\Delta \in smd(add((K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{E}) \boxtimes \mathcal{G}[-m, m])^{\star n})$$

Applying the exact functor $R\mathrm{pr}_{2,*}$ and observing that

$$R\mathrm{pr}_{2,*}\left((K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{E}) \boxtimes \mathcal{G}\right) = R\Gamma(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{E}) \otimes_k \mathcal{G}$$

by Derived Categories of Schemes, Lemma 08EU we conclude that

$$K = R\mathrm{pr}_{2,*}(L\mathrm{pr}_1^* K \otimes^{\mathbf{L}}_{\mathcal{O}_{X\times X}} \mathcal{O}_\Delta) \in smd(add(R\Gamma(X, K \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{E}) \otimes_k \mathcal{G}[-m, m])^{\star n})$$

The equality follows from the discussion in Example 0FYV. Since K is perfect, there exist $a \leq b$ such that $H^i(X, K)$ is nonzero only for $i \in [a, b]$. Since X is proper, each $H^i(X, K)$ is finite dimensional. We conclude that the right hand side is contained in $smd(add(\mathcal{G}[-m + a, m + b])^{\star n})$ which is itself contained in $\langle \mathcal{G} \rangle_n$ by one of the references given above. This finishes the proof. □